

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Safety and Effectiveness of Trastuzumab Emtansine (T-DM1) as Therapy in Chinese Participants With HER2 Positive Advanced Breast Cancer
Short Title/Acronym: T-DM1 Real-World Study in Chinese Advanced Breast Cancer
Protocol Number: ML44675 **NCT Number:** NCT05945927
Posting ID: 254787010413275641 **Sponsor Name:** Hoffmann-La Roche (Roche)
Investigational Product: Trastuzumab emtansine (T-DM1)
Phase of Development: Not Applicable **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To investigate the safety profile (Percentage of Adverse Events [% AEs]) of Trastuzumab Emtansine (T-DM1) as therapy in Chinese participants with HER2-positive advanced breast cancer in real-world clinical practice.
Secondary Objectives: To evaluate the real-world effectiveness of T-DM1, including real-world progression-free survival (rwPFS), time to treatment discontinuation (TTD), and treatment patterns (% of T-DM1 by line of therapy, dosage, dose intensity, and duration).

2.2 Background & Rationale To address the need for real-world evidence on the use of T-DM1 in the Chinese population. While T-DM1 is an approved antibody-drug conjugate (ADC) for HER2-positive advanced breast cancer, this prospective, post-marketing study investigates its safety and effectiveness specifically in Chinese patients who have previously been treated with a taxane and trastuzumab, administered according to routine clinical practice.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Observational, Post-marketing
Control Type: Single-arm **Blinding:** Open-label **Randomization:** N/A (Non-Probability Sample)

3.2 Planned Enrollment & Duration
Target N\$: 178 **Number of Sites:** Multicenter across China (e.g., Anyang Tumor Hospital, Cancer Hospital Chinese Academy of Medical Sciences, West China Hospital of Sichuan University, etc.)
Estimated Study Start: 13-Sep-2023 **Trial Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Female patients, Age ≥ 18 years. 2. Diagnosed with HER2-positive, unresectable locally advanced or metastatic breast cancer before the start of T-DM1 treatment. 3. Had prior treatment for breast cancer which must contain a taxane and trastuzumab. 4. Getting the treatment of T-DM1 for the first time after the diagnosis of breast cancer.

4.2 Exclusion Criteria 1. Patients not receiving treatment for HER2-positive breast cancer with T-DM1 according to standard of care and in line with the current summary of product characteristics (SPC)/local labeling. 2. Has been previously treated with T-DM1 before current clinical visit. 3. Currently participating in any clinical trials. 4. Previously participated in any clinical trials investigating anti-HER2 drug within 1 year of the initiation of T-DM1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Drug Details: Trastuzumab emtansine (ATC Code: L01FD03) **Dosage/Frequency:** Trastuzumab emtansine administered as per local clinical practice and local labeling. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Until disease progression, unacceptable toxicity, or discontinuation as determined by the treating physician in routine clinical practice.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant medications allowed as per routine standard of care. **Prohibited:** Concurrent participation in other clinical trials; recent participation (within 1 year) in investigational anti-HER2 drug trials.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Per routine care	Informed Consent, Medical History, Confirmation of HER2+ status and prior therapies
Baseline	Day 0	First Dose of T-DM1, Baseline clinical assessments
Interim	Per routine care	Safety Monitoring, Adverse Event tracking, Effectiveness Assessments (rwPFS)
End of Study	Follow-up End	Final Vital Signs, Exit Interview, Treatment discontinuation records (TTD)

(Note: As an observational study, all visits and clinical assessments strictly follow the local standard of care cadence rather than a rigid, protocol-mandated schedule).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants with Adverse Events (% AEs) to evaluate the safety profile of the treatment. **Measurement Method:** Clinical assessment and safety reporting recorded in the real-world setting.

7.2 Secondary & Exploratory Endpoints - Real-world Progression-Free Survival (rwPFS) - Time to Treatment Discontinuation (TTD) - Percentage of Participants Exposed to T-DM1 by Line of Therapy

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collected via routine clinical follow-ups and HCP interviews during standard care visits.

Serious Adverse Events (SAEs): Reported according to standard post-marketing pharmacovigilance regulations (typically within 24 hours). **Data Monitoring Committee (DMC):** Internal safety monitoring managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Real-World Safety Set and Effectiveness Set (Intent-to-Treat equivalent for observational data). **Sample Size Justification:** Sample size of 178 reflects real-world enrollment of eligible participants treated with T-DM1 at participating sites across China. **Handling of Missing Data:** Standard handling for observational real-world evidence (e.g., Multiple Imputation or complete-case analysis where appropriate).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP), local real-world evidence generation guidelines, and the Declaration of Helsinki.

Monitoring Plan: Data verification of electronic medical records (EMR) against eCRFs per observational study guidelines. **Audit Trail:** Details on eCRF locking and data clarification forms will be maintained by the sponsor's data management team.

11. Study Identifiers & Governance

Primary ID: {{ML44675}} **Registry Number:** {{NCT05945927}} **Posting ID:** {{254787010413275641}} **Sponsor/Lead Organization:** {{Hoffmann-La Roche / Roche}} **Study Title:** {{A Study to Evaluate the Safety and Effectiveness of Trastuzumab Emtansine (T-DM1) as Therapy in Chinese Participants With HER2 Positive Advanced Breast Cancer}} **Official Regulatory Title:** {{An Observational Study to Evaluate

the Safety and Effectiveness of Trastuzumab Emtansine (T-DM1) as Second- or Later-Line Therapy in Chinese Patients With HER2 Positive Advanced Breast Cancer}}

12. Clinical Framework (Breast Cancer Specific)

Primary Condition: {{HER2-Positive Advanced Breast Cancer}}
Diagnosis Threshold: {{Unresectable locally advanced or metastatic breast cancer, pre-treated with taxane and trastuzumab}} **Medical Methodology:** {{Real-world clinical practice / Standard of Care}} **Optimization Target:** {{Trastuzumab emtansine (T-DM1) targeted therapy delivery}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical oncology pathway for advanced breast cancer}} **Education Protocol (HCP):** {{Per standard local clinical practice and product labeling}} **Education Protocol (Patient):** {{Per standard local clinical practice}} **Quality Audit Mechanism:** {{Real-world data (RWD) collection & pharmacovigilance tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Follow-up until disease progression, unacceptable toxicity, or treatment discontinuation}} **Onsite Visit Cadence:** {{Determined by routine standard of care}} **Remote Monitoring Frequency:** {{Determined by routine standard of care}} **Checklist Review Interval:** {{Determined by routine standard of care}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				